where F represents the forward price. This simple extension of Black-Scholes is known as *Black's formula*. Black's formula is often used to value LIBOR caps and floors as well as swaptions, as discussed in Chapters 4 and 5.

A.6 Option Sensitivities

Second in importance only to the value of options themselves, the question of how option prices will change if the underlying asset price changes, if volatility changes, or as time passes is of paramount importance to an option trader. Think about it: if we're running an options book, we're probably going to want to know what happens to the value of our book if implied volatility increases. If we're net short options, this could hurt us. If we're net long options, an increase in implied volatility will likely mean an increase in the value of our option portfolio.[25]

The buzzwords that we'll be throwing around in this section are the so-called *greeks*: delta, gamma, vega, and theta, to name a few. Each of these provides a measurement of the change in the value of an option, given a change in an underlying variable on which the option price depends (while holding the other variables constant). It is important to note that the computation of these option sensitivities depends crucially on the particular model we employ. For example, if we are using the Black-Scholes model in which the underlying asset is assumed to be lognormally distributed, options sensitivities may be quite different from those computed using a different model.[26] Depending on the particular option model being employed, it may be rather easy and convenient to compute the various option sensitivities analytically. If not, we can always resort to numerical approximations, as we will discuss.

A.6.1 Delta

The first option sensitivity that we'll discuss (which is probably also the one that is most near and dear to an option trader's heart) is the change in the option price with respect to a change in the underlying asset price. This is known as the *delta* of the option. The reason that option traders are so concerned with delta is because once an option is traded, movement in the underlying can potentially change the value of the position dramatically.

Denote by $V(S, X, r, T, \sigma)$ the value of any option as a function of the underlying asset price, strike, risk-free rate, time to expiry, and implied volatil-

[25] Not all options increase (decrease) in value as implied volatility increases (decreases), as we will discuss later in this appendix.

[26] For example, in Chapter 5 we discuss the normal model as an alternative to Black-Scholes for the pricing of interest rate options.

ity. For ease of notion, we will often write V as a function of just one of these variables, with the implicit understanding that the option value actually depends on all of the parameters. The delta of the option is given by

$$\Delta = \frac{\partial V}{\partial S}. \tag{A.37}$$

For any price of the underlying asset, S, delta can be approximated by:

$$\Delta \approx \frac{\Delta V}{\Delta S} = \frac{V(S+\epsilon) - V(S)}{\epsilon}, \tag{A.38}$$

where $V(S)$ denotes the value of the option when the underlying is at S and ϵ is a small positive number. When should we use Equation A.37 to calculate delta versus using Equation A.38? Typically, people use Equation A.37 whenever it is analytically easy to compute that partial derivative, and if not, then they resort to the approximation in Equation A.38. Equation A.38 makes it easy to calculate the value of delta by computing the value of the option when the underlying is equal to $S + \epsilon$ and again when the underlying is equal to S. The difference of these two values divided by ϵ gives a close approximation to the true delta when ϵ is very small.

For European calls and puts on a non-dividend-paying asset, delta is easily determined analytically using the Black-Scholes formula:

$$\Delta_c = \frac{\partial c}{\partial S} = N(d_1) > 0 \tag{A.39}$$

$$\Delta_p = \frac{\partial p}{\partial S} = N(d_1) - 1 < 0. \tag{A.40}$$

Figures A.10 and A.11 show the delta of a call option and a put option, respectively, according to the Black-Scholes formula as a function of the underlying asset price.

But what does this really mean? When an option trader buys or sells an option, probably the first thing that he is worried about is his delta. Once that option is traded (and assuming the trader doesn't or can't immediately get out of the position), if the underlying moves around, it is going to affect the value of his option, often in a big way. Yes, while it is true that the trader's position has exposure to other things, too, such as movements in implied volatility, the thing that is often most likely to move in the near term and affect the value of his position is the underlying.

Delta hedging refers to trading in the underlying in conjunction with a position in options such that the changes in the net value of the portfolio are reduced as the underlying moves.[27] When delta hedging, it is often necessary to

[27] Again, this is not to suggest that other variables that drive the value of the option, such as implied volatility, are not moving. But delta hedging refers to immunizing changes in an option position due solely to movements in the underlying.

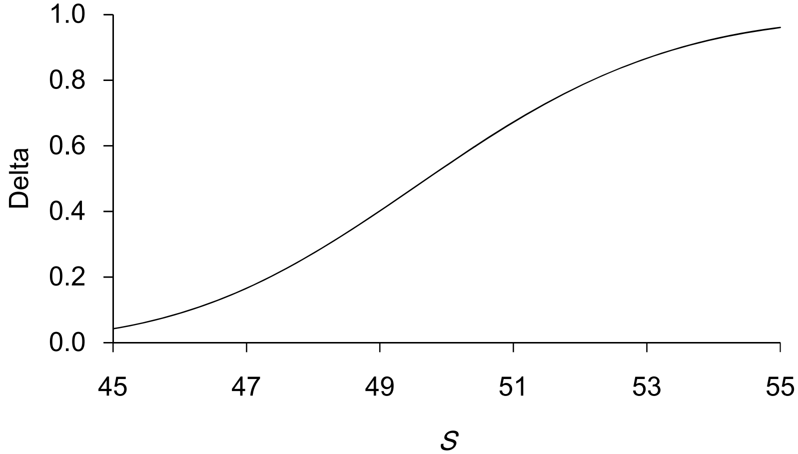

Figure A.10: Delta of Call Option as a Function of Underlying Asset When $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

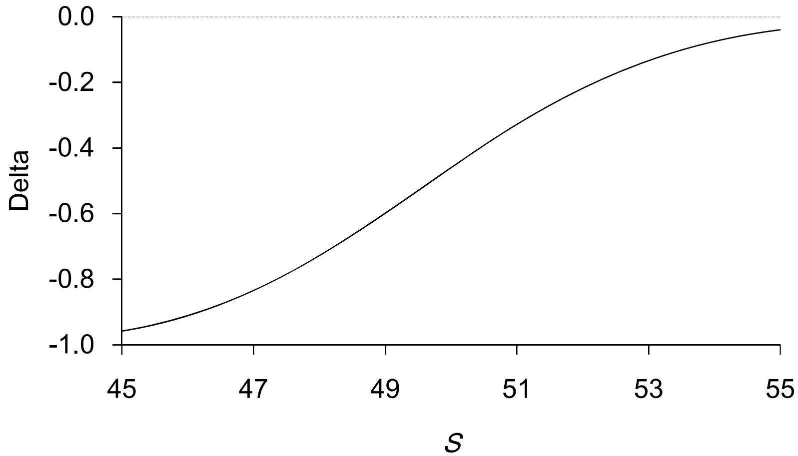

Figure A.11: Delta of Put Option as a Function of Underlying Asset When $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

readjust the amount of the underlying held in the portfolio since the delta of an option is not a constant but, rather, moves as a function of the price of the underlying and other variables that determine the price of the option. This process of adjusting the delta hedge is referred to as *dynamic delta hedging.*

To continue with this thought a bit more, suppose that an option trader sells an option and wants to delta hedge the position by trading in the underlying. The value of the position will be

$$-V + nS, \tag{A.41}$$

where n is the number of shares of the underlying held in the portfolio. A positive value of n indicates the trader is long the underlying, and a negative value indicates he is short. The change in the value of this portfolio with regard to a change in the price of the underlying is found by taking the partial derivative of the position in Equation A.41 and is

$$-\frac{\partial V}{\partial S} + n, \tag{A.42}$$

since of course $\partial S/\partial S = 1$. Oftentimes when delta hedging, an option trader is interested in maintaining a *delta-neutral* position, which is to say that the change in the value of his portfolio in Equation A.41 is zero for a small change in the underlying. In order to make this position delta neutral, we must set the expression in Equation A.42 equal to zero. Thus, in order to set up a delta-neutral hedge for an option we have sold, we set the position in the underlying

$$n = \frac{\partial V}{\partial S}. \tag{A.43}$$

By definition $\partial V/\partial S$ is Δ, the delta of the option. Thus, in order to maintain a delta-neutral position, we must set n equal to Δ. Inasmuch as the delta of an option is not a constant, an option trader must continuously adjust the amount of his hedge if he wants to maintain a delta-neutral position.

Note that if an option trader is instead long an option and wants to delta hedge by trading in n shares of the underlying, the value of his position is

$$+V + nS. \tag{A.44}$$

In order to maintain a delta-neutral position against this long option position, the trader will want to maintain a position in the underlying

$$n = -\frac{\partial V}{\partial S}. \tag{A.45}$$

The only difference between the formulation in Equation A.44 in which it is assumed the option trader is long the option and in Equation A.41 in which the option trader is short the option is the sign in front of the V, indicating the appropriate position. Once again, a positive quantity for n in Equations A.44

and A.45 indicates a long position, whereas a negative quantity indicates a short position.[28]

As an example, consider now the particular case of an option trader who sells a call option and wants to maintain a delta-neutral position. Assuming the trader employs the Black-Scholes model, then the trader will, in accordance with Equations A.39 and A.43, purchase $n = N(d_1)$ shares of the underlying. Inasmuch as the quantity $N(d_1)$ is positive, this means that the option trader who sold the call will buy the underlying to delta hedge. As a second example, consider the case of an option trader who buys a put option and wants to maintain a delta-neutral position. From Equation A.45, we see that the trader should set $n = -\partial V/\partial S$. Under the Black-Scholes model, this means that $n = -(N(d_1) - 1)$, which is positive by virtue of Equation A.40. This means that an option trader who delta hedges a long position in a put option will buy the underlying.[29]

A.6.2 Gamma

Gamma is difficult to conceptualize. Maybe it's because it's a second derivative. I don't know. Gamma measures the change in the delta of a derivative that occurs when the underlying asset price changes. More formally, gamma is given by

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}. \tag{A.46}$$

Gamma can be approximated by making use of the delta of an option as follows:

$$\Gamma \approx \frac{\Delta(S + \epsilon) - \Delta(S)}{\epsilon}, \tag{A.47}$$

where $\Delta(S)$ represents the delta of the derivative when the underlying asset price is S and ϵ is a small positive number.

For European calls and puts on a non-dividend-paying asset, gamma as per Equation A.46 is easy to determine analytically from the Black-Scholes formula:

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[28] Be careful in Equation A.45. The negative sign in this equation does not imply necessarily that n is negative; if $\partial V/\partial S$ is negative, then indeed n would be positive.

[29] There is nothing mysterious here. If we are long a put option, then as the underlying goes down, the value of our put goes up. If we want to remain delta neutral, we will have to be long some of the underlying. As the underlying goes down, losses in the value of the hedge will offset gains in the value of the option.

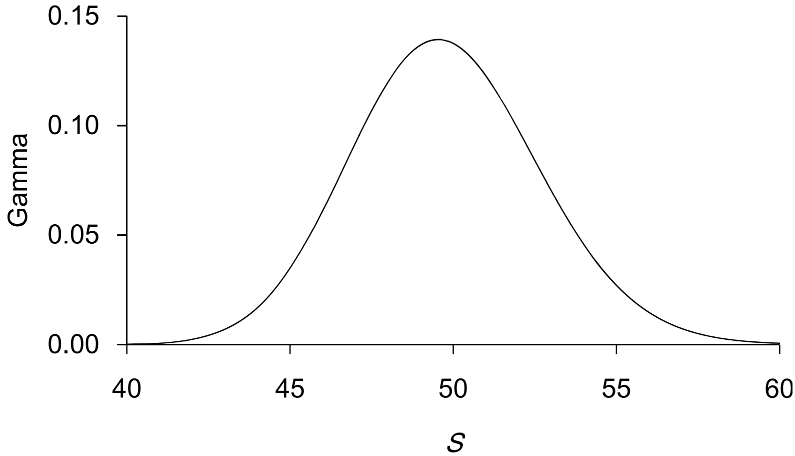

Figure A.12: Gamma of a Call Option as a Function of Underlying Asset When $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

$$\Gamma_c = \frac{\partial \Delta_c}{\partial S} = \frac{n(d_1)}{S\sigma\sqrt{T}} > 0 \tag{A.48}$$

$$\Gamma_p = \frac{\partial \Delta_p}{\partial S} = \frac{n(d_1)}{S\sigma\sqrt{T}} > 0, \tag{A.49}$$

where $n(\cdot)$ is the standard normal density function. Figure A.12 shows the gamma of a call option according to the Black-Scholes formula. As seen in this figure, gamma is largest when the underlying is near-the-money (meaning the underlying asset price is close to the strike) and falls off precipitously if the option is either deep in-the-money or deep out-of-the-money. If the option is deep in-the-money, then an option trader who is long the option and delta hedging is short nearly a full share of the underlying to remain delta-neutral. As the underlying moves around slightly, the option remains deep in-the-money, and the trader will not need to adjust the hedge much, meaning gamma is low. Similarly, if the option is deep out-of-the-money, an option trader who is long the option and delta hedging will be short almost none of the underlying. A small change in the underlying means the option will still be deep out-of-the-money and will not necessitate changing the delta hedge much, again meaning that gamma is low. It is when the underlying is around the strike that a change in the underlying implies the largest changes in the delta, which is to say that the gamma of the option is largest.

Consider now Figure A.13 that shows an option with positive gamma. Suppose we are long this option and are delta hedging it. The current value

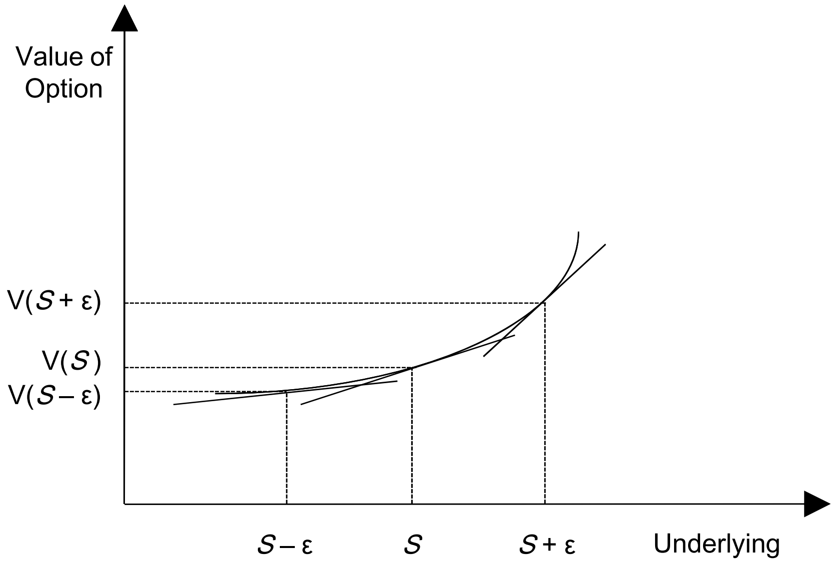

Figure A.13: An Option with Positive Gamma

of the underlying asset is S. In order to be delta neutral, our delta hedge is to short a number of units of the underlying asset equal to the slope of the tangent line to the option at S.[30] Let's suppose that we are going to dynamically delta hedge this option in the following manner: if the underlying price moves by an amount ϵ where $\epsilon > 0$, then we are going to readjust the amount of underlying we are short in order to once again make our portfolio delta neutral. Suppose now that the underlying trades up to $S + \epsilon$. Since the slope of the tangent line to the option price is steeper (i.e., the option price is now more sensitive to changes in the underlying), this means that the delta of the option is greater, and we will sell some more of the underlying at $S + \epsilon$. If the underlying subsequently falls back down to S, now the delta

[30]How do we know that our delta hedge is going to be short the underlying just by looking at this picture? From the picture it is clear that as the underlying goes up, the value of the option goes up (and as the underlying goes down, the value of the option goes down). Thus, in order to be delta neutral we will need to be short some of the underlying, as an increase in the underlying will cause our short to fall in value (and a decrease in the underlying will cause our short to rise in value). More formally, from Equation A.45, we know that our position in the underlying required to be delta neutral against our long in the option is $n = -\partial V/\partial S$. Since $\partial V/\partial S$, the slope of the tangent line at S, is positive, this means that n is negative.

has gone back down to where it was originally, and we need to pare back on the amount of the underlying we are short. Whereas we had just sold some underlying when it was trading at $S + \epsilon$, we now need to buy some back at S. If the underlying trades down to $S - \epsilon$, now the delta of the option (as indicated by the flattest of the three tangent lines) is even smaller, and we need to buy some more underlying, which we will do at the current price of $S - \epsilon$. And if the underlying trades back up to S, we will sell some more of the underlying when we adjust our delta hedge.

But look at what is happening here in terms of our position in the underlying. When we first buy the option, we start out shorting some of the underlying in order to have a delta-neutral position. When the underlying trades up, we sell some more. When it comes down, we buy some back at a cheaper price. When it comes down some more, we buy some more back at an even cheaper price. And when it subsequently goes back up, we wind up selling some at a more expensive price than we had just bought it. In other words, we are in a situation in which we keep buying low and selling high, which is a very good thing. The more that the underlying keeps going up and down, the more we can keep profiting by trading in the underlying, buying it at lower prices than at which we sold it, and selling it at higher prices than at which we bought it.[31]

This is the beauty of gamma. Being *long gamma* (which means the gamma of a position is positive) allows us to benefit from trading in the underlying if we are dynamically delta hedging.[32] Of course, this benefit comes at a price in the case of calls and puts: we have to shell out the purchase price of the option upfront in order to buy the option, which creates a long gamma position for us. If the actual volatility in the underlying is sufficiently large enough over the life of the option, then we can earn a substantial profit from buying the option and delta hedging. On the other hand, if the

[31] Even if we buy the option and set up a delta-neutral hedge by shorting some of the underlying, and the underlying subsequently moves only in one direction, the gamma of the option will benefit us. For example, suppose we buy the option depicted in Figure A.13 and the underlying subsequently moves from S to $S + \epsilon$ and sits there. In this case the value of the option will increase to a greater extent than will the associated loss in the hedge; the value of the call will go up to $V(S+\epsilon)$ and therefore benefit from the curvature of the price function, whereas the hedge will only go down linearly.

[32] When we are long gamma and delta hedging, our delta hedging activities can actually serve to stabilize the price movements in the underlying, assuming theses trading activities in the underlying are significant enough. As the price of the underlying goes up and we sell it to delta hedge, this tends, all else equal, to bring the price of the underlying back down. And as the price of the underlying goes down and we buy it in order to delta hedge, this exerts upward pressure on the price of the underlying. We will talk further about this potential stabilizing (or destabilizing, if we are short gamma) dynamic later on in this appendix when we consider binary options.

actual volatility of the underlying is low, we could lose money by buying the option and delta hedging.[33]

If we are *short gamma*, meaning the gamma of our position is negative, then the situation is reversed. As the underlying trades up, we need to buy some of the underlying in order to remain delta neutral, and as the underlying trades down, we need to sell some of the underlying to remain delta neutral.[34] Thus, if we are short gamma and are delta hedging, we tend to buy the underlying when it is high and sell it when it is low, which is not an advantageous thing to be doing in and of itself.[35] Inasmuch as we acquired the short gamma position by selling an option, we have taken in a premium upfront that offsets our potential losses from delta hedging. The actual profit or loss from selling an option and delta hedging will depend on the magnitude of actual price movements in the underlying and the frequency with which a trader readjusts his delta hedge.

The dynamics just described explain why many traders buy and sell short-dated options. While it is true that some short-dated option traders trade these options to take a view on the direction of the underlying and never delta hedge, many others are interested in trading these options for

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[33]The actual profit or loss from buying an option and delta hedging depends very much on the particular delta hedging strategy employed by the option trader. An option trader must make a decision both on the frequency and the minimum price movements in the underlying upon which he will adjust his delta hedge. Some option traders regard the process of making these choices as much of an art as a science. Question 10 in the problem set at the end of Chapter 5 allows us to consider this a bit further.

[34]Why is this happening? Let's consider a couple of examples to build some intuition. First, suppose we are short a call option, and we are delta hedging to remain delta neutral. We are short gamma, as per Equation A.48. In accordance with Equation A.43, we are going to set our delta hedge to $n = \partial c/\partial S = \Delta_c$, which is positive by virtue of Equation A.39. Thus, our delta hedge against our short call is to buy shares of the underlying. If the underlying trades up, the delta of the call option increases, and we need to buy more shares to remain delta neutral. If the underlying trades down, the delta of the option decreases, and we need to sell some shares of the underlying to remain delta neutral.

As a second example, suppose we are short a put option, and we are delta hedging to remain delta neutral. We are short gamma, as per Equation A.49. We are going to set our delta hedge to $n = \partial p/\partial S = \Delta_p$ in order to remain delta neutral. Note that this is a negative number according to Equation A.40. This means that when we sell the put, in order to set up a delta-neutral hedge we will sell shares. Now as the underlying trades up, the delta of the put increases (the delta is negative and becomes less negative, getting closer to zero), and we need to reduce our short by buying some shares to remain delta neutral. If the underlying trades down, the delta of the put decreases (the delta, which was negative, becomes more negative), and we have to sell some more shares of the underlying to remain delta neutral.

[35]In the case of an option position with negative gamma, the delta hedging activities of an option trader can actually serve to exacerbate movements in the underlying. As the underlying goes up and the option trader buys some of it, this will tend to push the price up further. And as the underlying goes down and the trader sells some to delta hedge, this serves, on the margin, to push the price down even further.

their gamma profile. An option trader who thinks that the *actual* (also called *realized*) volatility in the underlying will be significant enough so that potential profits from delta hedging will outweigh the cost of an option will be inclined to buy the option. One who expects low realized volatility in the underlying may be inclined to sell short-dated options, taking in premium, and hope for minimal costs associated with delta hedging the option. Oftentimes (but not always), sell-side option traders (i.e., dealers) who trade short-dated options with buy-side clients (who often don't delta hedge but are rather trading the options to take a directional view on the underlying) will delta hedge their option positions.[36]

A.6.3 Vega

The *vega* of an option is the change in the value of the option given a change in implied volatility. Formally, vega is given by

$$\frac{\partial V}{\partial \sigma}. \tag{A.50}$$

Vega can be approximated as follows:

$$\frac{V(\sigma + \epsilon) - V(\sigma)}{\epsilon}, \tag{A.51}$$

where $V(\sigma)$ represents the value of the derivative when the implied volatility is σ. For European calls and puts on a non-dividend-paying asset, vega is found from the Black-Scholes formula using Equation A.50:

$$\frac{\partial c}{\partial \sigma} = Sn(d_1)\sqrt{T} > 0 \tag{A.52}$$

$$\frac{\partial p}{\partial \sigma} = Sn(d_1)\sqrt{T} > 0. \tag{A.53}$$

[36]For those who find gamma a hard concept to get one's hands around, it may be easier to guess if an option trader is long or short gamma, given his demeanor. If the market for the underlying is moving around a lot, and you see an option trader who is running a delta-neutral book who looks like he's on the verge of a nervous breakdown, it's a pretty good bet that he's short gamma. It's hard to miss: that irascible look with eyes glued to the screen and teeth clenched as the underlying continues to move. Ask that guy a question, and you're likely to get a response like Charlie Sheen's character Bud Fox gave to Marv, played by John C. McGinley, in the movie *Wall Street* when Marv asked what was up and Bud screamed at him and told him to do his own homework. On the other hand, find an option trader who is also running a delta-neutral book but who has a care-free look on his face when the market for the underlying is moving and barely looks on the screens to see what the market is doing; and when he does look, you see a slight self-assured grin. That guy is most likely long gamma.